Automatic Radiology Report Generation: A Comprehensive Review and Innovative Framework

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***Abstract*— scientific research has consistently sought to improve human life quality, with a particular emphasis on advancing healthcare in hospitals and clinics. This study focuses on the development of intelligent systems to improve healthcare by assisting with disease diagnosis and treatment recommendations. Given the precision necessary in the medical area, we look at advanced technologies, specifically deep learning (DL) algorithms that use neural networks (NN) to simulate complex decision-making processes. We present a detailed review of the most recent breakthroughs in Automatic Radiology Report Generation (ARRG) systems, covering the many frameworks and approaches utilized in their implementation, as well as evaluating the most often used datasets and metrics. Additionally, we investigate the relevance of large language models (LLMs) in improving the development and interpretation of radiological reports. A proposed framework, detailing the best practices, future research directions, and implementation strategies, is also presented. This study aims to offer a thorough understanding of ARRG systems, benefiting researchers and practitioners in the field.**

Keywords—Healthcare, Automatic Radiology Report Generation, Deep Learning Algorithms, Neural Networks, Large Language Models.

I. Introduction

Scientific research has always aimed to improve human life, especially by advancing healthcare in hospitals and clinics. Researchers have developed intelligent systems to aid doctors in diagnosing illnesses, suggesting treatments, and making decisions in complex cases. Handling medical data with precision is crucial due to the rigorous and complex nature of the field. This has driven the adoption of advanced technologies, particularly machine learning algorithms, in creating intelligent medical systems. Deep learning algorithms, a subset of ML and a branch of artificial intelligence, use neural networks inspired by the human brain to simulate complex decision-making. Neural network research began in the 1950s [1] with Frank Rosenblatt's introduction of the perceptron algorithm. While early progress was limited by computational power, today's advances have removed these constraints, allowing deep learning to be widely applied across sectors like healthcare. DL shows great promise in improving diagnostic accuracy and treatment outcomes.

Technological developments in the field of deep learning over the last few years have given rise to new technologies in the health sciences. The computational burden and complexity of processing medical data such as X-ray images (X-rays, CT scans, mammograms, ultrasounds, MRIs) and textual data is reduced. As a result, radiology reports are automatically generated from X-ray images using deep learning algorithms. Deep learning is the foundation of automatic radiology report generation (ARRG).

ARRG was motivated by previous research on using the relationships between language and visual features to generate text descriptions of natural images. Initially, automated image captioning models had not been extended to medical images. These images are more challenging to handle due to their inherent similarities. Research on ARRG began in 2015 [2], when Shin and al. trained a Convolutional Neural Network (CNN) on CT scan and MRI images, followed by their pioneering study in this area in 2016 [3]. Thanks to the development of deep learning, researchers have introduced and improved a wide range of methods for optimising ARRG systems. In this study, we aim to achieve several key objectives:

- We plan to offer a comprehensive overview of the latest advancements in ARRG systems.
- We will examine various frameworks, network architectures, and enhancement techniques employed in the implementation of ARRG systems.
- We will evaluate the datasets and metrics commonly used for assessing these systems.
- We propose a theoretical framework that outlines future directions for research and potential implementation strategies.

This research aims to assist scholars and practitioners in understanding the overall workflow of ARRG systems and the diverse deep learning methods applied in this field. Our work is structured into six sections to systematically achieve our goal.

In Section 2, we first review the state of the art in Automatic Radiology Report Generation (ARRG). To provide a basic understanding, we describe the main process of ARRG systems in Section 3. The methods used to implement these systems are covered in Section 4, along with the various techniques and technologies utilized to create ARRG systems. Section 5 discusses the datasets and assessment metrics employed to test and evaluate ARRG systems. In Section 6, we introduce a proposed theoretical framework, detailing its conceptual model, potential benefits, and future application prospects. Following this, we offer an analytical analysis of the methodologies covered in our study and provide recommendations for future research and the practical implementation of ARRG systems.

II. Related works

1) Background

Radiological tests, like X-rays and MRIs, offer detailed insights into a patient's body without needing invasive surgery. These exams require radiologists to write detailed reports that various healthcare specialists can use. Imagine a radiological report as the main conversation thread between a radiologist and other healthcare professionals it's that vital! This report carries crucial information for diagnosing and managing patient care in hospitals. While there isn't one-size-fits-all structure for these reports, the European Society of Radiology (ESR) [4] emphasizes that they must be clear, correct, concise, complete, consistent, and coherent. Interestingly, despite the lack of a universal standard, many countries have adopted their own methods for structuring radiological reports. The ESR elucidated these methods in their 2023 paper on structured reporting in radiology [5], and Nobel, J. Martijn et al. discussed them further

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in their systematic review [6]. For decades, this traditional way of producing reports from radiographic images has dominated, especially in countries less advanced in digital and new technologies. However, this method has shown its limitations on numerous occasions. Writing radiology reports is a task specifically reserved for radiologists, who are the only experts in the field. Despite the growing number of specialists in medical radiology, the gap between radiologists and patients remains wide. This gap translates into long waiting times for patient treatment. What's more, the reports provided may contain errors or be subject to subjective interpretation. Beyond the human element, the increasing quantity and variety of radiological images is prompting technology research specialists to explore the automation of the reporting process. The aim is to optimise the performance of hospitals in terms of healthcare, ensuring that patients are treated more accurately and more quickly.

2) Can an ARRG system be set up via LLMs?

Thanks to advancements in AI, especially deep learning, large language models (LLMs) have become one of the most promising technologies. LLMs are foundation models trained on vast amounts of data, allowing them to understand and generate natural language and other content types, performing a wide range of tasks. As mentioned earlier, LLMs are built on deep learning, particularly transformers, which have grown significantly in recent years. The work by [7] offers a thorough understanding of LLM architectures, their application fields, and highlights some open issues and future research directions. Humza Naveed et al. provide a comprehensive overview of research in the field of LLMs in their paper [8]. One sector where LLMs hold significant potential is healthcare. Due to the critical nature of healthcare, any system implemented in this domain must exhibit high performance. Given the impressive results demonstrated by LLMs, their potential for the automatic generation of radiological reports is worth exploring. Studies conducted on this topic have shown promising results. Papers [9], [10], discuss various aspects of LLM-based models in ARRG systems. According to these studies, LLM-based models in ARRG systems encounter several challenges, including:

- Lack of accuracy: Ensuring the precision of radiological reports is difficult. Any inaccuracies can lead to misdiagnoses and sometimes-improper treatment, even if LLMs have shown remarkable capabilities in language understanding and generation.

- Privacy concerns: a major concern when handling medical data. It is crucial to protect the confidentiality and security of patient information, especially when implementing LLMs in healthcare systems.

- Hallucination: LLMs have the capability to produce reasonable yet inaccurate information. In radiology, this poses a risk of generating reports that could potentially mislead healthcare professionals.

- Need for large training datasets: training LLMs well requires extensive datasets. In medical domains, assembling such datasets while maintaining patient privacy and data diversity can be challenging.

- Necessity of human review: Despite advancements in automation, the generated reports often need to be reviewed and evaluated by human radiologists to ensure accuracy and reliability.

3) Challenges

Setting up a system for automatically generating radiology reports involves overcoming a number of different obstacles. One of the first difficulties encountered by researchers in this field is the extraction of key features from the dataset. These feature extraction challenges arise primarily from a large patient population, the quality of radiology images and the heterogeneity of diseases affecting different organs, as highlighted by the Royal College of Radiologists (RCR) in their report in 2022 [11]. In addition to the extensive size of radiological data, the complexity of modalities such as MRI and CT images presents a significant issue. Following the extraction of information from these images, the next challenge is generating clear and understandable reports. Research into the automatic generation of radiological reports is one of the most recent fields. This is a specific application of automatic image captioning (AIC). Similarly, there are models of ARRG systems based on non-deep learning (DL) models [12], which are more complex to implement. These ARRG systems require manual feature specification in order to develop mathematical and statistical models for pattern recognition. Feature design and extraction require specific knowledge, which is not universally available.

Recent advances in deep learning techniques and image captioning have significantly contributed to the development of ARRG systems. One of the pioneering works in this field was conducted by Shin et al. [2] in 2015. In 2016, Shin et al. proposed another foundational work on ARRG [3]. Since 2018, there has been a notable surge in work on this topic [13], [14], [15]. In these studies, researchers have explored various deep learning methods to implement ARRG systems, some of which are more complex or accurate than others. Several review articles, [16], [17], [18] provide insights into the primary objectives of ARRG systems. some literature reviews focus on specific categories of medical images. For instance, [19] offers a comprehensive survey of ARRG systems utilizing chest radiographs.

III. ARRG General Process

The implementation of ARRG systems can be divided into different phases, each intended to accomplish specific tasks with the goal of developing a system that successfully meets the prospects of its developers.

Generally, the workflow for implementing ARRG systems is similar for both DL and non-DL approaches. Initially, the process starts with the input of a type medical image, then, pre-processing it to enhance its quality and suitability for analysis. Next, feature extraction step is performed to identify images' key features that are relevant aimed to detecting anomalies.

In this step, DL techniques (particularly CNNs), are commonly used. Non-DL approaches [12] employ classical machine learning algorithms for tasks such as classification or clustering of the dataset, utilizing methods like K-means and cosine similarity. The next phase is report generation.

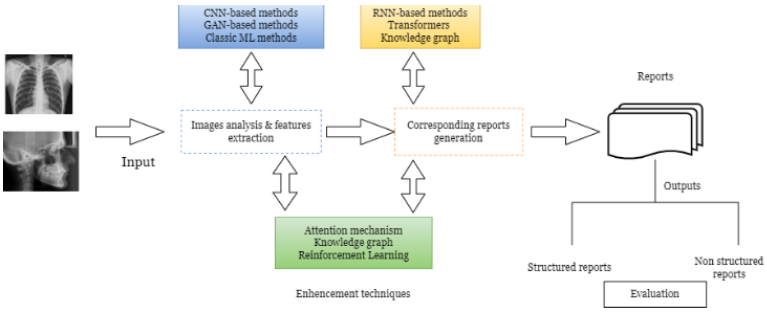

Figure 1: General workflow for ARRG systems

In DL models, the extracted features are processed using another DL model, classically based on a RNN or a transformer architecture. This model is trained on existing radiology reports to learn the language patterns and structure employed by radiologists. The generated report can be either structured or unstructured. Finally, the AI-generated report should go in a crucial evaluation phase. A radiologist reviews the report to verify the findings, add their expertise and context, and finalize the report for the patient's medical record. This step ensures the accuracy and reliability of the ARRG system. Overall, many works employ various methods to produce optimal ARRG systems, regularly combining DL with other techniques to create a robust model capable of meeting all the expectations of ARRG systems. As illustrated in Figure 1, this integration pulls the strengths of each method to enhance the overall performance of the ARRG system.

IV. Using Deep Learning approaches for ARRG systems

As mentioned, DL approaches have significantly advanced the field of ARRG systems. In this section, we investigate numerous approaches. We explore the different network architectures used, the

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frameworks that support these systems, and the enhancement techniques applied to further improve their performance.

1) Network architectures

Researchers working on ARRG systems adopt various DL network architectures to improve the performance of their models, each of these architectures contributing exclusively to the effectiveness and accuracy of ARRG. In this sub-section, we will explore these specific architectures, and describe how they may enhance the capabilities of ARRG systems.

a) Convolutional Neural Network (CNN)

Extracting features from an input image is the most crucial step in ARRG systems. Indeed, we need to capture the main features in a given medical image to obtain as much information as possible. It takes an input image and represents it in a reduced dimensionality space, allowing the model to focus on the most relevant information for the task. CNNs [20] are feedforward neural networks with convolution layers that operate on image input to extract features such as shape, edges and textures. It performs extremely well in the case of spatial data such as images and is thus preferably used in ARRG systems as visual encoder. CNNs are comprised of three types of layers. These are convolutional layers, pooling layers and fully-connected layers [20]. The convolutional and pooling layers perform feature extraction, whereas a fully connected layer maps the extracted features to the final output, such as a categorical output. The CNN architecture is used essentially in image analysis field. That architecture is almost used by all ARRG system, [13], [14], [21] , to extract features from radiology images.

b) Recurrent Neural Network (RNN)

After features are extracted from radiology images, we need to translate those features into naturel sentences. The most common network architecture used for this task is the Recurrent Neural Network (RNN) [22]. The concept of RNNs was first proposed by Rumelhart et al. [23], in a 1986 letter published by Nature. They described a new learning procedure using a self-organizing neural network. RNNs are adept at capturing sequential information within input data and share parameters across different elements in the input stream [19]. In ARRG systems, RNNs are often paired with CNNs and handle the textual decoding process. Several variations of RNN architectures exist, with the most popular being: SimpleRNN, LSTM, GRU, bidirectional RNN, and deep RNN. There are two modified branches of CNN-RNN architectures which are CNN-HRNN[1] [13], [21] and CNN-SRNN[2] [14]. These two architectures are found in large researches on ARRG systems due their high performance on previous works.

c) Transformers

This architecture is essentially based on the encoder-decoder framework. Transformers were initially introduced for natural language processing (NLP) tasks but have since been adopted in broader deep learning fields. They rely on a technique called attention [24], which measures the relationships between pairs of input tokens. These tokens can be words in text strings or parts of images for visual transformers. This architecture exclusively uses self-attention units [24]. Attention, a technique used in computer science, mimics the human brain's ability to focus on relevant parts of the input [23]. To understand sequential data clearly, we need to consider the context in which a word or sentence is used. This concept of context motivated the introduction of the attention mechanism in 2015 by Bahdanau et al. [25]. Their attention module allows the model to focus on the parts of a sentence that are relevant to predicting the next word In ARRG systems, transformers are typically connected to a CNN encoder [26], [27], [28]. This fusion joins the strengths of both architectures, leveraging CNNs for robust feature extraction from radiological images and Transformers for context-aware textual generation. This approach enhances the capability of ARRG systems to produce accurate and clinically meaningful reports based on complex medical imaging data.

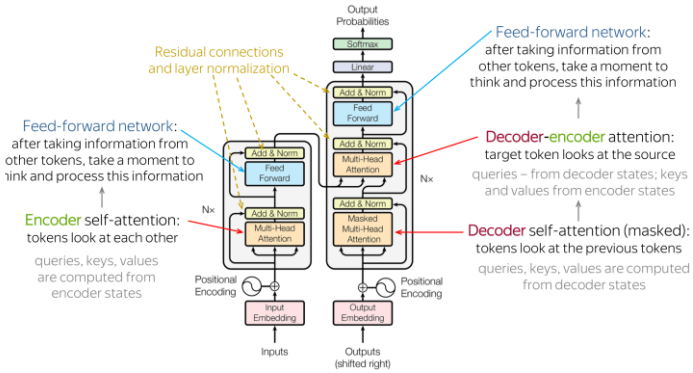

Figure 2: Transformer architecture[3]

d) Generative Adversarial Networks (GANs)

Generative adversarial networks GANs are a unique type of neural network architecture. They consist of two parts: a generative network that creates solutions to a task, and a discriminative network that compares and rates the generated solutions against known real data [23]. Unlike CNNs, GANs aim to train a generator network that can produce new data that closely resembles a given training dataset. Ian Goodfellow et al. introduced the first work on GAN architecture in 2014 [29]. While not as commonly used in ARRG as CNNs and RNNs, GANs have been applied in some research for the segmentation of spinal structures in lumbar spine MRI scans [17].

2) Frameworks

There are essentially three categories of frameworks used for implementing ARRG systems (Figure 6).

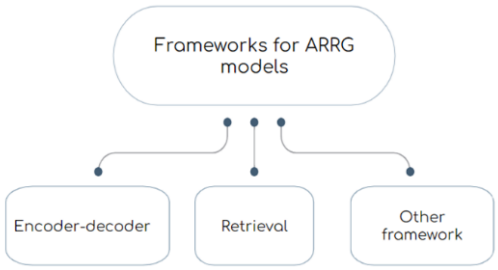

Figure 3: ARRG models frameworks

a) Encoder-decoder

The encoder-decoder architecture is the most implemented architecture in recent researches on ARRG systems. This is due to the good results achieved by the precursors in this field. It's basic framework developing DL-based for AIC [30] model. It was firstly used for sequence-to-sequence generation and later for image to sequence generation. The encoder-decoder architecture is based mainly on two components. A visual encoder that extracts image features and a textual decoder, which learns the mapping from the image representation to text representation and consequently generates sequences. The encoder-decoder use CNN architectures for visual features extraction and RNN are used as textual decoder.

[1] Hierarchical Recurrent Neural Network
[2] Sequential Recurrent Neural Network
[3] https://lena-voita.github.io/nlp_course/

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Research in this framework architecture [13], [14], [21], [31] shown good results that explains the lot interest in this framework architecture. In the same path, Yaowei Li and al. [32] have addressed the issue of ARRG systems through an encoder-decoder framework which contains three novel modules: Latent Space Unifier (LSU), Cross-modal Representation Aligner (CRA), and Text-to-Image Refiner (TIR). They exploit a discrete variational autoencoder (dVAE) in the LSU component. Bhanu Prakash Voutharoja and al. [33] propose a framework to learn discriminative image and report features by distinguishing them from their closest peers, i.e., hard negatives. They use an encoder to extract image representation and passes it to a decoder to generate a report.

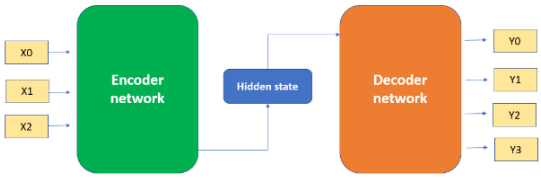

Figure 4: Encoder-decoder architecture[4]

b) Retrieval

The retrieval frameworks are the least used in ARRG systems. Unlike architectures with fixed structures, these frameworks design retrieval methods that match extracted image features to corresponding sentence templates. These frameworks have two main group of methods. First group methods focus on the visual features extraction from the given medical image and the second one refers to the construction of template database for retrieval. During their study on systematic review for ARRG systems, [17] identified four researches, [12], [14], [15], [34] which applied retrieval architecture to build their ARRG systems. In [34], the cosine similarities between the test embedding and all the training embeddings in the database are calculated and the most similar exam is retrieved from the database. Its diagnostic text is then assigned to the test exam. This work faced challenges due to high diversified dataset and the lack of consideration for patient history, which is not considered by most publicly available medical images datasets. [15], treated sentence selection in ARRG systems as a multi-label classification problem using knowledge graph and retrieves methods. Jianmo Ni and al. [12] approach the task as a cross-modal retrieval method. They propose a model that measures the similarity between images and abnormal findings, and identifies fine-grained relevant image regions for each abnormal finding. The RadBERT model [35] is used for each report to obtain its nearest neighbour. For each report, a cosine similarity score identifies its nearest neighbour

c) Other

Some researchers [36], [37] addressed challenges in ARRG systems by transforming them into a multi-label classification problem. The frameworks used in this case can be divided into two main groups, one of which utilizing CNN-based methods while other uses Generative Adversarial Network GAN-based methods for classification accompanied by image segmentation.

3) Enhancement techniques

To achieve the best possible results in ARRG systems, field specialists often add additional deep learning techniques into their base architectures to enhance system performance.

a) Attention mechanism

Bahdanau and al. [25] introduced this technique in 2015. Attention is an interface connecting the encoder and decoder that provides the decoder with information from every encoder-hidden state. With this technique, the model is able to selectively focus on valuable parts of the input sequence and hence, learn the association between them. Attention mechanism was first proposed in the NLP field and still actively researched in the field. A general overview of attention mechanism is given by [38], [39] during their review work on attention mechanism. In ARRG systems, attention mechanism is used to improve visual features extraction or to capture more semantic information on input data. There are several types of attention mechanism implemented in ARRG according to the specific needs of searchers. Attention mechanism technique is used in encoder-decoder frameworks [13].

b) Graph structure

Graph explicitly represents relationships between entities. These entities are represented as nodes and edges. We can encode both image and text as a graph. In ARRG, Graph Neural Networks (GNNs) such as the Graph Convolutional Network (GCN) and Graph Transformer (GTR) have been utilized to leverage the graph structure. They are typically trained on prior knowledge of radiology ontology and constructed from corresponding reports [17]. Very promising results have been obtained using these structure graphs [15], [40].

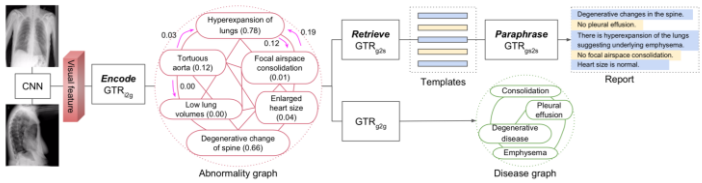

Figure 5: ARRG system architecture using knowledge-driven encode, retrieve and paraphrase module [15]

c) Reinforcement learning

Reinforcement learning is a feedback-based machine learning technique in where an agent learns to behave in an environment by performing the actions and seeing the results of actions. For each successful action, the agent gets positive feedback, and for each unsuccessful action, the agent gets negative feedback or penalty. Reinforcement learning has achieved good results in sequence generation models such as visual captioning, text summarization, and machine translation. Reinforcement learning can be used to enhance learning and updating model parameters through policy gradients. This approach directly uses the evaluation metrics as a reward in ARRG [14].

V. Datasets & Evaluation Metrics

1) Datasets

To develop effective ARRG models in ARRG systems, searchers have to train and test their models with some medical data. This data primarily consists of radiology images and their corresponding reports. Before the advent of ARRG systems, there were a number of databases used to train and test deep learning models. These datasets can be visual or textual (see Table 1). The first medical dataset used in ARRG research is the OpenI dataset [41]. It contains 7,470 chest X-rays of lateral and frontal view out of 32, 717 unique patients. The OpenI dataset contains also 3,955 radiology reports. Demner-Fushman and al in 2016 [42], proposed IU X-ray. In total, 3996 radiology reports were collected from the Indiana Network for Patient Care and 8121 associated images from

[4] https://vitalflux.com/encoder-decoder-architecture-neural-network/

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the hospitals' picture archiving systems. The final dataset contains 7470 Chest (lateral and frontal view) X-rays and 3,955 radiology reports. In 2019, [43] introduced a new dataset called Medical Information Mart for Intensive Care-Chest X-ray (MIMIC-CXR). It contains 371,920 chest X-rays linked to 227,943 reports. The IU X-ray and MIMIC-CXR datasets are among the most widely used datasets in ARRG research.

Dataset	**Images**		**Reports**		
	Image modality & body part (nO. images)	*Format*	*Type*	*number*	*Report format*
[41] OpenI dataset	Chest (lateral and frontal view) X-rays (n=7, 470)	DICOM & PNG	NA	3,955	XML
[44] liver CT	Liver (3D CT images with liver masks and bounding boxes) CT (n=50)	MAT	Structured reports	50	RDF
[42] IU X-ray	Chest (lateral and frontal view) X-rays (n=7, 470)	JPEG, DICOM	Semi Structured reports + MeSH tags	3,995	TXT
[45] ChestX-ray8	Chest (frontal view) X-ray (n=108,948)	DICOM	Semi structured + labels	NA	TXT
[46] ChestX-ray14	Chest (frontal view) X-rays (n=112,120)	DICOM	Semi structured + labels	NA	
[47] CheXpert	Chest (frontal and lateral) X-ray (n=224,316)		Free text reports + labels	NA	TXT
[43] MIMIC-CXR	Chest (Frontal and lateral views) X-ray (n=377,110)	JPEG, DICOM	Semi Structured report + labels	227,835	TXT

Table 1: Datasets used in ARRG systems

2) Evaluation metrics

To evaluate the performance of reports generated by ARRG systems, several metrics are employed. Most metrics used for ARRG evaluation were originally developed for NLP. These metrics can be divided into two main groups: natural language evaluation metrics (NLEMs) and clinical efficacy metrics (CEMs) (see Table 2). NLEMs are used to check the clearness and fluency of the generated reports [19]. NLEMs may include BLEU [48], ROUGE [49], CIDEr [50], METEOR [51], SPICE [52], and MIRQI [40]. BLEU, ROUGE, and METEOR metrics were originally designed to evaluate machine translation and summarization tasks. In contrast, CIDEr, SPICE, and MIRQI were specifically developed for evaluating general image descriptions. On the other hand, clinical efficacy metrics (CEMs) are used to ensure the clinical relevance and accuracy of the generated reports. Examples of CEMs include Keywords Accuracy (KA) [13], Medical Subject Heading (MeSH) Accuracy (MA) [53], and Clinical Accuracy (CA) [54]. MIRQI can also be used as a CEM [40].

Metrics	**Purpose**	**Description**	**Algorithm**	**Strengths**	**Weaknesses**
Machine translation and summarization metrics					
BLEU_2002 [48]	Machine translation	A precision-based metric that counts how many n-grams from the generated text exist among ground truth references	n-gram precision	correlation with human judgment	Lack of explicit word matching
ROUGE_2004 [49]	Document summarization	A recall-based metric that counts how many n-grams from the ground truth references exist in the generated text	n-gram recall	works well in single document summarization	can't be applied in multi-document summarization
Images description metrics					
CIDEr_2015 [50]	image captioning	Image description evaluation using human consensus	n-gram with corpu reweighting	Works in linguistics means	May weight irrelevant sentence's details
SPICE_2016 [52]	image captioning	graph encodes objects, attributes and relationships between words of the image's caption	fobjects*fattributes*frelations	Can match noun / object between captions	Reliant on the performance of parsing

Table 2: Evaluation metrics used to evaluate ARRG

VI. Our Proposed Framework for ARRG

1) Framework Overview

In this study, we review several ARRG studies. We first present a comprehensive overview of the latest advancements in ARRG systems including the use of Large Language Models (LLMs) and their potential role in ARRG systems. Secondly, we provide an

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overview of the General ARRG Process. Then, we examine various frameworks, network architectures, and enhancement techniques employed in the implementation by ARRG models. We summarized as well as the datasets and the evaluation metrics commonly used in these models. Moreover, we proposed a new practical ARRG framework that outlines future directions for research and potential implementation strategies. Finally, we discuss the current challenges and future research directions of ARRG. Overall, ARRG shows promise in playing a crucial role in clinical workflows, but further advancements are needed to produce reports that are as accurate as those written by radiologists

2) Conceptual Model

Our proposed framework is divided into several modules, each of which performs a specific role throughout the implementation process. Each module corresponds to a stage or sub-stage of the implementation process with specific methods and algorithms.

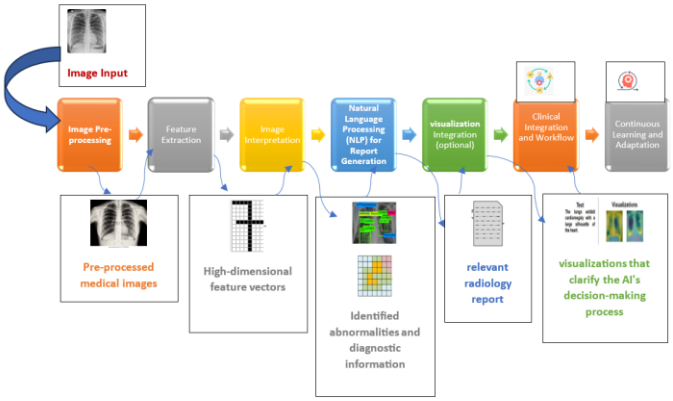

Figure 6: Our proposed framework

a) Image Pre-processing

The input for this step is raw medical images (e.g., X-rays, CT scans, MRIs). These images are then cleaned using techniques and methods like noise reduction, normalization, segmentation, and data augmentation [55]. These techniques and methods filter and normalize the images to standardize intensity values (noise reduction, normalization). There have been many advancements in this field. Linwei Fan et al., in their paper [56], provide an overview of some techniques used to denoise images. To obtain high-quality images, segmentation methods are used in the pre-processing step. In medical imaging, this involves the precise delineation of anatomical structures and regions of interest (ROIs) from imaging data. Two main models can be used for image segmentation. The first is the U-Net model, a type of convolutional neural network (CNN) architecture initially designed for biomedical image segmentation, introduced in 2015 [57]. The second is the Mask R-CNN model, an advanced object detection framework that extends Faster R-CNN by adding a branch for predicting segmentation masks on each Region of Interest (RoI). The first paper on the Mask R-CNN model dates back to 2017 [58]. Teerath Kumar et al., in their paper [55], provide a general overview of data augmentation, these techniques include rotation, scaling, flipping, and elastic deformations.

At the end of this process, the output consists of pre-processed and augmented images ready for feature extraction.

b) Feature Extraction

After the image cleaning process, the next step is feature extraction. This module takes the pre-processed and augmented images as input and produces high-dimensional feature vectors representing the content of the images. At this stage, deep learning-based algorithms like EfficientNet [59], a family of convolutional neural networks (CNNs) developed by Google AI, can be used to achieve better performance in terms of accuracy and computational efficiency compared to previous models.

c) Image Interpretation

The main goal of this step is to identify abnormalities and diagnostic information within the feature vectors from the feature extraction module. There are two possible approaches: using only CNN algorithms or a hybrid approach that combines CNNs with other architectures to improve the interpretation model. The only CNN architecture in this module includes Classification and Detection. EfficientNet [59] and SE-ResNet [60] can be used for classification tasks. For detection algorithm, object detection models like YOLOv5 [61] can be applied. The third component of the CNN architecture is a segmentation model such as U-Net [57] or DeepLabv3+ [62] (an advanced image segmentation model designed for high accuracy in segmenting objects within images).

d) Natural Language Processing (NLP) for Report Generation

Diagnostic information and features from the image interpretation module are taken as input, and coherent, clinically relevant radiology reports are produced as output through this module. End-to-End Neural Networks can be applied in this part of the framework.

e) Visualization Integration

In this module, XAI[5] techniques can be used to make the decisions and predictions of ARRG AI systems transparent and understandable to humans. Attention visualization or gradient-based methods are used to explain model decisions. This module takes as input the generated reports and decision-making data from the interpretation module and produces explanations and visualizations that clarify the AI's decision-making process. Additionally, this module includes also explanations along with the generated reports for review by radiologists.

f) Clinical Integration and Workflow

ARRG AI-generated reports and feedback from radiologists are taken as input at this stage of the framework, and finalized radiology reports integrated into clinical workflows are produced as output. During this module, radiologists review, edit, and approve AI-generated reports. Interoperability with PACS[6] and RIS[7] for seamless integration is also ensured at this step of the ARRG system implementation. The user interface must be user-friendly for interaction and feedback, while maintaining compliance with healthcare regulations.

g) Continuous Learning and Adaptation

This final module takes as input radiologist feedback and new data, and then we update and improve the model's performance over time. The aim of the system's performance is the enhancement by integrating feedbacks from radiologists and incorporating new data. Consequently, the system performance should monitor and evaluated using both quantitative and qualitative metrics. This ensures the development of a high-performance system that

[5] Explainable Artificial Intelligence
[6] Picture Archiving and Communication System
[7] Radiology Information System

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effectively addresses the challenge of automatic generation of radiology reports

Discussion and perspectives

A much progress has been made in developing ARRG systems. Using advanced deep learning algorithms, especially neural networks have proven to significantly improve diagnostic accuracy and efficiency in ARRG. Reviewing different frameworks and approaches highlights how adaptable and versatile these systems are in handling various types of radiological data. Regarding the generation of the radiological reports, adding large language models also boosts their ability to generate more detailed and relevant reports. Due to general image/caption datasets are different from radiology image/report datasets, using regular AIC models for ARRG might create reports that look real but are not accurate. Therefore, it's important to customize DL approaches specifically for ARRG. To summarize, the DL approaches are the most used for setting up an ARRG system. This is undoubtedly due to the latest advances in deep learning, but also to its promising results. Among the works that have adopted this approach, there are several architectures that use a variety of techniques and algorithms, some of which are more optimal than others. The architecture most commonly used with this approach is the CNN-RNN. There are two main variations of this architecture, namely the CNN-HRNN and the CNN-SRNN. As presented in [13], the ARRG system based on the encoder-decoder framework, have shown great result during tests. In addition, researchers use some networks architectures to enhance their models. These architectures are mainly: attention mechanism, transformers, knowledge graphs and reinforcement learning. Although not the most common approaches, non-deep learning methods are still being explored by researchers in this field. These methods are considered arduous and complex to implement due to the modularity of their components and the need for strong engineering skills to combine the required algorithms. For retrieval-based methods, they use the similarity of data features, which can reduce repetitive sentences and lessen the bias toward generating normal findings [12], [14], [15]. Studying hybrid models that combine CNNs with other types of neural network architectures could greatly enhance the accuracy and reliability of ARRG systems. For Datasets, in this review, we argue that only the MIMIC-CXR dataset meets three key criteria: it is large-scale, publicly available, and contains original reports. However, future research should aim to create and use standardized datasets and evaluation metrics for ARRG systems. This will help ensure consistent benchmarking, making it easier to compare and improve results across different studies. For evaluating the generated reports, many ARRG models were assessed using standard AIC metrics like BLEU, ROUGE, METEOR, and CIDEr. These metrics focus on language fluency and are based on n-gram overlap. However, n-gram overlap alone does not necessarily ensure that two sentences convey the same meaning [48], [49]. It was widely believed that accurately detecting diseases should be more important than language fluency when evaluating generated reports. Therefore, various clinical efficacy metrics were created, focusing on the accuracy, precision, and recall of disease labels for the ARRG task [53], [54]. However, these metrics might not be enough to fully evaluate a report because describing diseases also involves qualifiers, certainties, and negations, not just specific labels. In the future, researchers should study how to combine different types of data (like radiological images, patient history, and lab results) to make the generated reports more comprehensive and accurate.

VII. Conclusion

In this study, we review several ARRG studies. We first present a comprehensive overview of the latest advancements in ARRG systems including the use of Large Language Models (LLMs) and their potential role in ARRG systems. Secondly, we provide an overview of the General ARRG Process. Then, we examine various frameworks, network architectures, and enhancement techniques employed in the implementation by ARRG models. We summarized as well as the datasets and the evaluation metrics commonly used in these models. Moreover, we proposed a new practical ARRG framework that outlines future directions for research and potential implementation strategies. Finally, we discuss the current challenges and future research directions of ARRG. Overall, ARRG shows promise in playing a crucial role in clinical workflows, but further advancements are needed to produce reports that are as accurate as those written by radiologists.

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